

Boss Challenge — Paper 52 (Class 7)

Light | Soil | Integers | Comparing Quantities

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Light from a torch reaches your eye along the shortest path because light travels in:
(A) a straight line
(B) only downward toward the ground
(C) a curved path that bends around corners
(D) a wavy zig-zag through air
2. The dark, crumbly material in topsoil, formed from rotted dead plants and animals, is called:
(A) gravel
(B) bedrock
(C) clay
(D) humus
3. The result of adding any integer to its additive inverse (its opposite) is always:
(A) a negative number
(B) the integer itself
(C) 0
(D) 1
4. Reduced to its lowest terms, the ratio 6 : 9 becomes:
(A) 3 : 2
(B) 1 : 2
(C) 2 : 3
(D) 6 : 9
5. An object 12 cm in front of a plane mirror forms its image. The distance between the object and its image is:
(A) 12 cm
(B) 24 cm
(C) 6 cm
(D) 0 cm
6. Reading a soil profile from the surface downward, the correct order of layers is:
(A) bedrock, then subsoil, then topsoil
(B) topsoil, then bedrock, then subsoil
(C) subsoil, then topsoil, then bedrock
(D) topsoil, then subsoil, then bedrock
7. The value of $(-7) + (+3)$ is:
(A) +4
(B) +10
(C) -10
(D) -4

8. Converted into a percentage, the fraction $\frac{3}{5}$ equals:

- (A) 60%
- (B) 53%
- (C) 35%
- (D) 30%

9. When a ray of light strikes a mirror, the angle of incidence is always:

- (A) equal to the angle of reflection
- (B) zero for every ray
- (C) always 90 degrees
- (D) twice the angle of reflection

10. The layer of a soil profile that is darkest and richest in humus, where most plant roots grow, is the:

- (A) subsoil
- (B) the water table
- (C) bedrock
- (D) topsoil

11. The product $(-5) \times (-4)$ equals:

- (A) -9
- (B) +20
- (C) +9
- (D) -20

12. What is 25% of the number 80?

- (A) 20
- (B) 40
- (C) 55
- (D) 25

13. A dentist uses a small mirror to see an enlarged image of a tooth. The mirror is:

- (A) concave
- (B) convex
- (C) plane
- (D) a flat piece of glass

14. The slow breaking down of rocks into fine particles by sun, water, wind and living things, which forms soil, is called:

- (A) evaporation
- (B) condensation
- (C) germination
- (D) weathering

15. A diver is 12 m below the sea surface, shown as -12 m. She rises 5 m. Her new position is:

- (A) -17 m
- (B) +7 m
- (C) -5 m
- (D) -7 m

16. A student scored 15 marks out of 25 in a test. As a percentage, the score is:

- (A) 60%
- (B) 40%
- (C) 75%
- (D) 15%

17. The wide-angle mirror on the side of a car shows a small, upright image and a large field of view. It is a:

- (A) concave mirror
- (B) a magnifying lens
- (C) convex mirror
- (D) plane mirror

18. Soil made of very fine, tightly packed particles that holds the most water and feels sticky when wet is:

- (A) gravelly soil
- (B) sandy soil
- (C) loamy soil
- (D) clayey soil

19. On a number line, which of these integers is the greatest?

- (A) -3
- (B) -10
- (C) -7
- (D) -20

20. In a concave mirror a 4 cm object forms a 12 cm image. The magnification, as the ratio image : object, is:

- (A) 4 : 12
- (B) 3 : 1
- (C) 8 : 1
- (D) 1 : 3

21. The word AMBULANCE is painted reversed on the front of the vehicle so that, in a driver's mirror, it reads correctly. This is because a plane mirror causes:

- (A) total reflection of colour
- (B) lateral inversion
- (C) magnification
- (D) dispersion

22. Through which soil does water drain (percolate) the fastest, because its large particles leave big air spaces?
- (A) loamy soil
 - (B) clayey soil
 - (C) black cotton soil
 - (D) sandy soil
23. When you simplify $5 - (-3)$, the result is:
- (A) 8
 - (B) 2
 - (C) -8
 - (D) -2
24. The simple interest on Rs 2000 at 5% per year for 2 years is:
- (A) Rs 2000
 - (B) Rs 2200
 - (C) Rs 100
 - (D) Rs 200
25. An image that can be caught on a screen, because rays of light actually meet there, is called a:
- (A) erect image
 - (B) real image
 - (C) virtual image
 - (D) diminished image
26. The soil best suited for growing most crops, because it holds enough water yet still drains and is rich in humus, is:
- (A) pure sandy soil
 - (B) gravelly soil
 - (C) loamy soil
 - (D) pure clayey soil
27. The temperature in a hill town was 4 deg C at noon and fell by 9 degrees by midnight. The midnight temperature was:
- (A) 13 deg C
 - (B) -13 deg C
 - (C) -5 deg C
 - (D) 5 deg C
28. A shopkeeper buys a pen for Rs 40 and sells it for Rs 50. The profit percentage is:
- (A) 50%
 - (B) 20%
 - (C) 25%
 - (D) 10%

29. Sunlight passing through a glass prism spreads into a band of seven colours. This splitting of white light is called:

- (A) lateral inversion
- (B) dispersion
- (C) magnification
- (D) reflection

30. In a percolation test, 200 mL of water takes 40 minutes to pass through a soil sample. The percolation rate (amount per minute) is:

- (A) 40 mL per minute
- (B) 0.2 mL per minute
- (C) 8 mL per minute
- (D) 5 mL per minute

31. Among all the negative integers, the greatest (largest in value) one is:

- (A) -100
- (B) 0
- (C) 1
- (D) -1

32. If 2 kg of apples cost Rs 50, then 5 kg of the same apples cost (by the unitary method):

- (A) Rs 150
- (B) Rs 125
- (C) Rs 100
- (D) Rs 250

33. Two plane mirrors are placed at right angles (90 degrees) to each other. The number of images of an object placed between them is:

- (A) 4
- (B) 2
- (C) infinite
- (D) 3

34. Paddy (rice) is grown in fields kept flooded with water. The most suitable soil for paddy is therefore:

- (A) loamy soil
- (B) sandy soil
- (C) gravelly soil
- (D) clayey soil

35. The product $(-1) \times (-1) \times (-1)$ equals:

- (A) +1
- (B) -1
- (C) +3
- (D) -3

- 36.** A surface reflects 3 out of every 5 units of light that fall on it. The percentage of light it reflects is:
- (A) 3%
 - (B) 35%
 - (C) 40%
 - (D) 60%
- 37.** A 3 cm tall object placed before a concave mirror gives an image 9 cm tall. The ratio of image height to object height (the magnification) is:
- (A) 3 : 1
 - (B) 6 : 1
 - (C) 1 : 3
 - (D) 9 : 1
- 38.** Earthworms are called a farmer's friend mainly because, as they burrow, they:
- (A) eat the roots of crops
 - (B) drink up all the soil water
 - (C) loosen the soil and let in air
 - (D) make the soil more acidic
- 39.** A cave floor is 8 m below ground level (-8 m) and a bird hovers 6 m above the ground (+6 m). The vertical distance between the bird and the cave floor is:
- (A) 14 m
 - (B) 48 m
 - (C) 2 m
 - (D) -2 m
- 40.** A jacket priced at Rs 800 is sold at a 10% discount. The discount amount is:
- (A) Rs 100
 - (B) Rs 10
 - (C) Rs 80
 - (D) Rs 720
- 41.** We are able to see this page, even though it does not glow, because it:
- (A) absorbs all the light falling on it
 - (B) produces its own light
 - (C) reflects light into our eyes
 - (D) bends light by refraction
- 42.** A soil sample of mass 100 g is dried and loses 18 g as water vapour. The percentage of water in the original sample was:
- (A) 18%
 - (B) 82%
 - (C) 36%
 - (D) 1.8%

43. The value of $(+6) \times (-3)$ is:

- (A) +18
- (B) -3
- (C) +3
- (D) -18

44. The decimal 0.35 expressed as a percentage is:

- (A) 0.35%
- (B) 35%
- (C) 350%
- (D) 3.5%

45. Reflection of light from a rough, uneven surface, which sends rays off in many directions, is called:

- (A) dispersion
- (B) diffused (irregular) reflection
- (C) regular reflection
- (D) total internal reflection

46. Listed from largest particle to smallest, the correct order is:

- (A) sand, gravel, clay, silt
- (B) clay, silt, sand, gravel
- (C) silt, clay, sand, gravel
- (D) gravel, sand, silt, clay

47. The value of $(-20) \div (+4)$ is:

- (A) -80
- (B) -16
- (C) -5
- (D) +5

48. A soil sample contains 9 mL of water in every 60 mL of soil. The percentage of water is:

- (A) 15%
- (B) 9%
- (C) 60%
- (D) 25%

49. A periscope, used to see over a wall, works using two plane mirrors fixed inside a tube at an angle of:

- (A) 45 degrees to the tube
- (B) 30 degrees to the tube
- (C) 0 degrees, lying along the tube
- (D) 90 degrees flat against the tube

50. The five main components found in a sample of fertile soil are minerals, humus, water, air and:

- (A) salt
- (B) plastic
- (C) acid
- (D) living organisms

51. Using the distributive rule, $(-3) \times (4 + (-6))$ equals:

- (A) 6
- (B) -6
- (C) -30
- (D) 30

52. Which is the larger ratio, 3 : 4 or 5 : 8?

- (A) 5 : 8
- (B) 3 : 4
- (C) cannot be compared
- (D) they are equal

53. A convex lens is thicker in the middle and bends parallel rays so that they:

- (A) turn back the way they came
- (B) change colour to red
- (C) converge to a point
- (D) spread apart and never meet

54. Dumping plastic bags, factory chemicals and excess fertilizer onto land leads to:

- (A) weathering
- (B) soil pollution
- (C) germination
- (D) percolation

55. The predecessor of -9 (the integer just before it) is:

- (A) -10
- (B) -8
- (C) 10
- (D) -9

56. If the simple interest earned is Rs 300 on a principal of Rs 2000, the amount (principal plus interest) is:

- (A) Rs 2000
- (B) Rs 2300
- (C) Rs 1700
- (D) Rs 300

57. In a rainbow the colours always appear in a fixed order, with the colour on the outer edge being:
- (A) violet
 - (B) green
 - (C) red
 - (D) white
58. Two soils are tested: through Soil X, 300 mL of water passes in 30 minutes; through Soil Y, 300 mL passes in 60 minutes. The ratio of their percolation rates (X : Y) is:
- (A) 1 : 1
 - (B) 30 : 60
 - (C) 1 : 2
 - (D) 2 : 1
59. The absolute value of -8, written $|-8|$, is:
- (A) 0
 - (B) 16
 - (C) -8
 - (D) 8
60. The ratio 2 : 5 expressed as a percentage is:
- (A) 40%
 - (B) 50%
 - (C) 20%
 - (D) 25%
61. Standing between two parallel plane mirrors facing each other, you see an endless line of repeating images. The number of images formed is:
- (A) exactly two
 - (B) infinite (very large)
 - (C) three
 - (D) none
62. Why is the hard, unbroken layer at the very bottom of a soil profile called bedrock the most difficult to dig through?
- (A) it has the most air spaces
 - (B) it is mostly water
 - (C) it is solid, unweathered rock not yet broken into particles
 - (D) it is full of soft humus
63. The result of dividing any non-zero integer by zero is:
- (A) 0
 - (B) infinity, as a whole number
 - (C) the number itself
 - (D) not defined

64. 60% of the 25 students in a class chose science as their favourite subject. The number of such students is:
- (A) 10
 - (B) 60
 - (C) 25
 - (D) 15
65. A 5 cm tall pencil is held upright before a plane mirror. The height of its image in the mirror is:
- (A) 10 cm
 - (B) 5 cm
 - (C) 2.5 cm
 - (D) 0 cm
66. On a hot afternoon, soil loses moisture because the Sun's heat causes the water in it to:
- (A) evaporate into the air
 - (B) turn into humus
 - (C) sink down to the bedrock
 - (D) freeze into ice
67. Arranged in ascending order (smallest first), the integers -2, 5, -8, 0 become:
- (A) 5, 0, -2, -8
 - (B) -2, -8, 0, 5
 - (C) -8, -2, 0, 5
 - (D) 0, -2, -8, 5
68. The price of a notebook rises from Rs 20 to Rs 25. The percentage increase is:
- (A) 50%
 - (B) 20%
 - (C) 5%
 - (D) 25%
69. Of every 100 units of light striking it, a certain shiny surface reflects 80 units. The percentage of light it reflects is:
- (A) 20%
 - (B) 800%
 - (C) 8%
 - (D) 80%
70. Farmers add manure and compost to their fields chiefly to:
- (A) to make the soil drain faster
 - (B) to turn the soil into bedrock
 - (C) replace humus and return nutrients to the soil
 - (D) to kill all the earthworms

71. Multiplying any integer by -1 has the effect of:
- (A) leaving it unchanged
 - (B) reversing its sign
 - (C) doubling its size
 - (D) making it zero
72. Two ratios are equivalent when they simplify to the same value. Which ratio is equivalent to 3 : 4?
- (A) 9 : 12
 - (B) 4 : 3
 - (C) 6 : 12
 - (D) 3 : 8
73. A concave mirror is used in a car's headlight and a torch chiefly to:
- (A) make the bulb last longer
 - (B) change the colour of the bulb
 - (C) gather light into a strong parallel beam
 - (D) spread the light in every direction
74. Compared with the hot surface on a summer day, the soil deep inside a cave stays:
- (A) exactly the same temperature
 - (B) cooler
 - (C) hotter than the surface
 - (D) always at freezing point
75. The successor of -1 (the integer just after it) is:
- (A) 0
 - (B) -2
 - (C) 1
 - (D) -1
76. Out of 100 units of light striking a window, 90 units pass through. The percentage of light transmitted is:
- (A) 10%
 - (B) 90%
 - (C) 100%
 - (D) 9%
77. A virtual image, such as the one you see in an ordinary plane mirror, differs from a real image because it:
- (A) is always upside down
 - (B) glows with its own light
 - (C) cannot be caught on a screen
 - (D) is always larger than the object

- 78.** The amount of water a soil can hold against gravity, before extra water drains away, is called its:
- (A) humus content
 - (B) particle size
 - (C) water-holding capacity
 - (D) percolation rate
- 79.** The sum $(-15) + (+15)$ is:
- (A) -30
 - (B) 0
 - (C) 15
 - (D) 30
- 80.** The percentage 40% written as a fraction in simplest form is:
- (A) $\frac{1}{4}$
 - (B) $\frac{2}{5}$
 - (C) $\frac{2}{4}$
 - (D) $\frac{4}{10}$
- 81.** A concave lens (thinner in the middle) always forms an image that is:
- (A) real, inverted and larger
 - (B) real and the same size
 - (C) virtual, erect and smaller than the object
 - (D) inverted but the same size
- 82.** Of every 50 mL of water poured onto a soil sample, 30 mL drains out the bottom. The percentage of water that drained through is:
- (A) 60%
 - (B) 40%
 - (C) 30%
 - (D) 20%
- 83.** The product of three negative integers, such as $(-2) \times (-3) \times (-1)$, is:
- (A) negative
 - (B) positive
 - (C) equal to 1
 - (D) zero
- 84.** A 50-litre water tank is 30% full. The volume of water in it is:
- (A) 15 litres
 - (B) 20 litres
 - (C) 35 litres
 - (D) 30 litres

85. The image formed by a plane mirror is best described as:
- (A) real, inverted and magnified
 - (B) virtual, upside down and smaller
 - (C) virtual, erect, same size and laterally inverted
 - (D) real, erect and the same size
86. Plants grown in pure clayey soil often grow poorly mainly because the soil:
- (A) contains far too much humus
 - (B) drains away water far too quickly
 - (C) holds too much water and lets in too little air
 - (D) has particles that are far too large
87. On a number line, starting at 0 and taking two steps of -3 each, you land on:
- (A) 6
 - (B) 0
 - (C) -3
 - (D) -6
88. If 5 pencils cost Rs 30, the cost of 8 such pencils is:
- (A) Rs 40
 - (B) Rs 56
 - (C) Rs 30
 - (D) Rs 48
89. Among a plane mirror, a concave mirror and a convex mirror, the one whose image is always smaller than the object is the:
- (A) concave mirror
 - (B) all three equally
 - (C) convex mirror
 - (D) plane mirror
90. Black soil, found in the Deccan region and excellent for cotton, is also known as:
- (A) laterite soil
 - (B) black cotton soil (regur)
 - (C) desert sandy soil
 - (D) alluvial soil
91. The additive identity, the integer that leaves any number unchanged when added, is:
- (A) the number itself
 - (B) -1
 - (C) 0
 - (D) 1

92. A magnified image is 7 times the height of the object. Written as a ratio, image height : object height is:

- (A) 7 : 7
- (B) 14 : 1
- (C) 7 : 1
- (D) 1 : 7

93. A ray of light hits a plane mirror so that its angle of incidence is 30 degrees. The angle between the incident ray and the reflected ray is:

- (A) 60 degrees
- (B) 90 degrees
- (C) 15 degrees
- (D) 30 degrees

94. Which property of topsoil most directly explains why it is the layer farmers most want to protect from erosion?

- (A) it contains the bedrock
- (B) it is the layer richest in humus and nutrients
- (C) it has the largest stones
- (D) it is the hardest layer to dig

95. Reading the integers on a thermometer, which is the LOWEST (coldest) temperature?

- (A) 0 deg C
- (B) -12 deg C
- (C) 3 deg C
- (D) -5 deg C

96. A class of 40 students has 24 girls. The percentage of girls in the class is:

- (A) 16%
- (B) 24%
- (C) 60%
- (D) 40%

97. Sunlight appears white, but a prism shows it is really made up of:

- (A) seven colours mixed together
- (B) no colour at all
- (C) only red and blue
- (D) a single pure colour

98. Before sowing seeds a farmer ploughs and loosens the soil mainly so that:

- (A) the soil turns into solid bedrock
- (B) air and water can reach the roots and roots can spread easily
- (C) all the humus is removed from the field
- (D) the soil becomes harder for water to enter

99. Subtracting, $(-4) - (-9)$ comes to:

- (A) 5
- (B) 13
- (C) -13
- (D) -5

100. If 20% of a number is 40, then the number itself is:

- (A) 60
- (B) 200
- (C) 400
- (D) 8